Non-Functional Requirements & Latency Numbers

Phase 1 · Fundamentals · Estimated time: 1.5 hours

1. Concept From First Principles

English:

Functional requirements describe what a system does. Non-functional requirements (NFRs) describe how well it does it: how fast, how available, how many concurrent users, how consistent. HLD interviews are graded far more on how you reason about NFRs than on the functional feature list.

Hindi:

Functional requirements बताते हैं कि एक system क्या करता है। Non-functional requirements (NFRs) बताते हैं कि वह कितना अच्छा करता है: कितना तेज़, कितना available, कितने concurrent users, कितना consistent। HLD interviews में feature list से कहीं ज़्यादा यह देखा जाता है कि आप NFRs पर कैसे सोचते हैं।

SLI, SLO, SLA -- the reliability vocabulary

English:

- SLI (Service Level Indicator): the raw, measured metric -- e.g., P99 latency of an endpoint.
- SLO (Service Level Objective): your internal target for that metric.
- SLA (Service Level Agreement): the external, often contractual promise to customers, usually looser than the SLO.

Hindi:

- SLI (Service Level Indicator): असली, मापा गया metric -- जैसे किसी endpoint का P99 latency।
- SLO (Service Level Objective): उस metric के लिए आपका internal लक्ष्य।
- SLA (Service Level Agreement): customers से किया गया बाहरी, अक्सर contractual वादा, जो आमतौर पर SLO से थोड़ा ढीला होता है।

Important Words:

- **Contractual** – अनुबंधित / समझौते वाला

Backend engineer analogy

English:

If you've ever set up an uptime monitor tracking p95 response time on an API, you were tracking an SLI. The alert threshold you configured was an informal SLO. A formal SLA is the same idea, written into a contract.

Hindi:

अगर आपने कभी किसी API के p95 response time को track करने वाला uptime monitor बनाया है, तो आप एक SLI track कर रहे थे। जो alert threshold आपने set किया, वह एक अनौपचारिक SLO था। एक formal SLA वही idea है, बस एक contract में लिखा हुआ।

Understanding percentiles (P50, P95, P99)

English:

Averages lie. If 99 requests take 50ms and 1 request takes 5000ms, the average looks fine but that one user had a terrible experience. Percentiles fix this: P99 means 99% of requests were faster than this value -- it captures your worst common case.

Hindi:

Averages धोखा दे सकते हैं। अगर 99 requests को 50ms लगे और 1 request को 5000ms, तो average ठीक लगेगा लेकिन उस एक user का experience बहुत खराब रहा। Percentiles इसे ठीक करते हैं: P99 का मतलब है कि 99% requests इस value से तेज़ थीं -- यह आपके सबसे बुरे आम मामले (worst common case) को दिखाता है।

Latency numbers every engineer should know (approximate)

Operation (English – reference table)	Approximate latency
L1 cache reference	~1 nanosecond
Main memory (RAM) reference	~100 nanoseconds
Redis / in-memory cache read (network)	~0.5-1 millisecond
SSD random read	~100 microseconds
Relational DB query (simple, indexed)	~1-10 milliseconds
Cross-region network round trip	~100-150 milliseconds

English:

You don't need to memorize exact numbers -- you need the relative intuition: memory is much faster than SSD, which is much faster than a database query, which is much faster than a cross-region network hop.

Hindi:

आपको सटीक numbers याद करने की ज़रूरत नहीं -- आपको एक relative समझ चाहिए: memory, SSD से बहुत तेज़ है, जो database query से बहुत तेज़ है, जो cross-region network hop से बहुत तेज़ है।

Important Words:

- **Intuition** – सहज समझ

English:

One more useful concept: an error budget. If your SLO is 99.9% availability over 30 days, that means you're implicitly allowed about 43 minutes of downtime that month -- that allowance is your error budget.

Hindi:

एक और उपयोगी concept: error budget। अगर आपका SLO 30 दिनों में 99.9% availability है, तो इसका मतलब है कि आपको उस महीने लगभग 43 मिनट का downtime implicitly allowed है -- वही allowance आपका error budget है।

Important Words:

- **Allowance** – अनुमत मात्रा

2. Real-World Example / Case Study

English:

Payment gateways like Razorpay publish explicit uptime SLAs to merchants because downtime directly costs merchants real revenue. Internally, their engineering teams track much stricter SLOs so normal operational hiccups don't breach the customer-facing SLA.

Hindi:

Razorpay जैसे payment gateways merchants को साफ-साफ uptime SLAs देते हैं क्योंकि downtime से merchants का असली revenue सीधे नुकसान में जाता है। अंदरूनी तौर पर, उनकी engineering teams कहीं सख्त SLOs track करती हैं ताकि आम operational दिक्कतें customer-facing SLA को ना तोड़ें।

Important Words:

- **Hiccup (figurative)** – छोटी अड़चन / रुकावट
- **Breach (verb)** – तोड़ना / उल्लंघन करना

English:

A second example: Noon's product search likely tracks a P95 or P99 latency SLO specifically, because a slow search response during a peak sale event directly costs conversions.

Hindi:

एक और उदाहरण: Noon की product search शायद खास तौर पर P95 या P99 latency SLO track करती है, क्योंकि किसी peak sale के समय धीमा search response सीधे conversions का नुकसान करता है।

Important Words:

- **Conversion (business)** – कन्वर्ज़न / खरीद में बदलना

3. Diagram (English labels kept as-is)

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Response times for 100 requests to /checkout (sorted):

  [ 50 requests: ~40-60ms ]  <- P50 sits here (median)
  [ 44 requests: ~60-200ms ]
  [  5 requests: ~200-400ms ] <- P95 sits here
  [  1 request:  ~4800ms   ]  <- P99 sits here (worst common case)

  AVERAGE of all 100  =~ 91ms   <- looks fine, HIDES the outlier
  P99                 =  4800ms <- reveals the real worst-case pain
```

4. Trade-offs Table (Reference Table – English)

Metric choice	What it shows	What it hides
Average (mean)	General central tendency	Outliers and worst-case pain
P50 (median)	The typical user experience	Everything the slower half experiences
P95 / P99	Near-worst-case experience	Requires more monitoring maturity

5. Common Interview Questions

Q1: "What's the difference between SLI, SLO, and SLA?"

English:

Model approach: SLI is the raw measurement, SLO is your internal target, SLA is the external contractual promise. Give one concrete example threading all three together.

Hindi:

Model approach: SLI असली माप है, SLO आपका internal लक्ष्य है, SLA बाहरी contractual वादा है। तीनों को जोड़ते हुए एक concrete उदाहरण दें।

Q2: "Why would you design around P99 latency instead of average latency?"

English:

Model approach: Explain that averages can look healthy while a meaningful fraction of users have a bad experience, and at scale, even a small P99 tail represents a large absolute number of frustrated users.

Hindi:

Model approach: समझाएं कि average भले ही ठीक दिखे, कई users का experience खराब हो सकता है, और बड़े scale पर एक छोटा सा P99 tail भी बहुत सारे परेशान users का बड़ा आंकड़ा बन जाता है।

Q3: "What is an error budget, and how would you use it to make a release decision?"

English:

Model approach: Define it as the allowed unreliability implied by your SLO. If it's nearly gone, pause risky changes; if there's budget left, move faster.

Hindi:

Model approach: इसे SLO से मिलने वाली allowed unreliability के रूप में define करें। अगर budget लगभग खत्म है, risky changes रोक दें; अगर budget बचा है, तो तेज़ी से आगे बढ़ें।

6. Practice Exercises

English:

- Given [20ms, 22ms, 25ms, 30ms, 4000ms] for 5 requests, calculate the average and identify the outlier.
- Write a one-sentence SLO for a checkout API's availability and one for its P99 latency.
- Explain why adding a Redis cache is likely to help more than optimizing the SQL slightly.

Hindi:

- [20ms, 22ms, 25ms, 30ms, 4000ms] इन 5 requests के लिए average निकालें और outlier पहचानें।
- checkout API की availability के लिए एक SLO और उसके P99 latency के लिए एक SLO लिखें।
- समझाएं कि Redis cache जोड़ना SQL को थोड़ा optimize करने से ज़्यादा क्यों मदद करेगा।

7. Curated YouTube Videos (Reference – English)

Language	Video & Channel	Why it helps
English	"SLO & SLI Explained: Service Level Objectives for Beginners"	Breaks down the SLI/SLO relationship.
English	"SLO vs SLI vs SLA vs Error Budget | Google SRE"	Ties vocabulary to real SRE practice.
Hindi	"What is SLA in Hindi | SLA kya hota hai"	Hindi explanation of SLA fundamentals.
Hindi	"Complete System Design Roadmap 2025" — CodeHelp	Covers NFRs in Hindi.

8. Key Takeaways

English:

- Non-functional requirements are what HLD interviews mostly grade you on.
- SLI = measured metric, SLO = internal target, SLA = external contractual promise.
- Percentiles (P50/P95/P99) reveal real user pain that averages hide.
- P99 represents your near-worst-case experience and is what production systems alert on.
- Know the relative order of latency: memory < SSD < DB query < network.
- An error budget ties reliability to a concrete, trackable number.

Hindi:

- Non-functional requirements ही वह चीज़ हैं जिन पर HLD interviews सबसे ज़्यादा focus करते हैं।
- SLI = मापा गया metric, SLO = internal लक्ष्य, SLA = बाहरी contractual वादा।
- Percentiles (P50/P95/P99) वह असली दर्द दिखाते हैं जो averages छुपा देते हैं।
- P99 आपके near-worst-case experience को दिखाता है और production systems इसी पर alert करते हैं।
- Latency का क्रम याद रखें: memory < SSD < DB query < network।
- एक error budget reliability को एक concrete, track किए जा सकने वाले number से जोड़ देता है।

General Words Glossary

A consolidated list of the important English words used throughout this chapter, with Hindi meanings and simple explanations — useful for revising vocabulary after finishing the chapter.

English Word	Hindi Meaning	Simple Explanation
Allowance	अनुमत मात्रा	The permitted amount of something
Breach (verb)	तोड़ना / उल्लंघन करना	To break or violate an agreement
Contractual	अनुबंधित / समझौते वाला	Legally or formally agreed upon
Conversion (business)	कन्वर्ज़न / खरीद में बदलना	A visitor turning into a paying customer
Hiccup (figurative)	छोटी अड़चन / रुकावट	A minor, temporary problem
Intuition	सहज समझ	A natural sense or understanding of something